

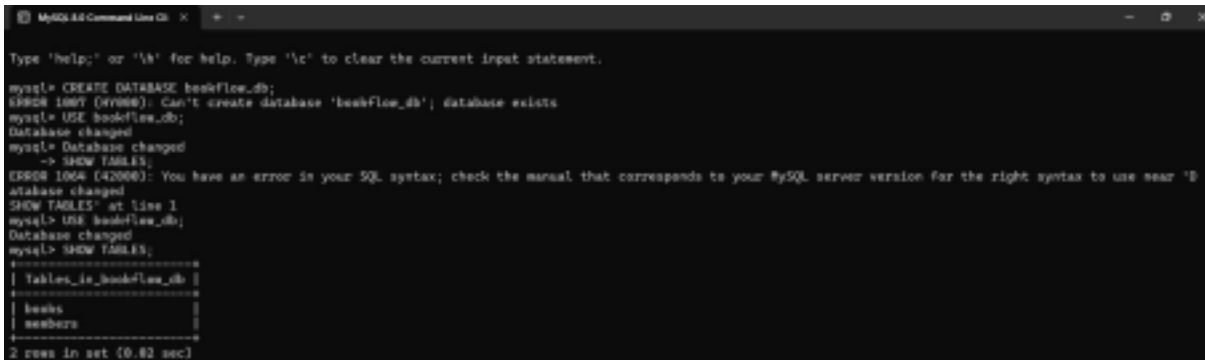
Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-1

Step 1 :Database Setup Query 1: Create the database

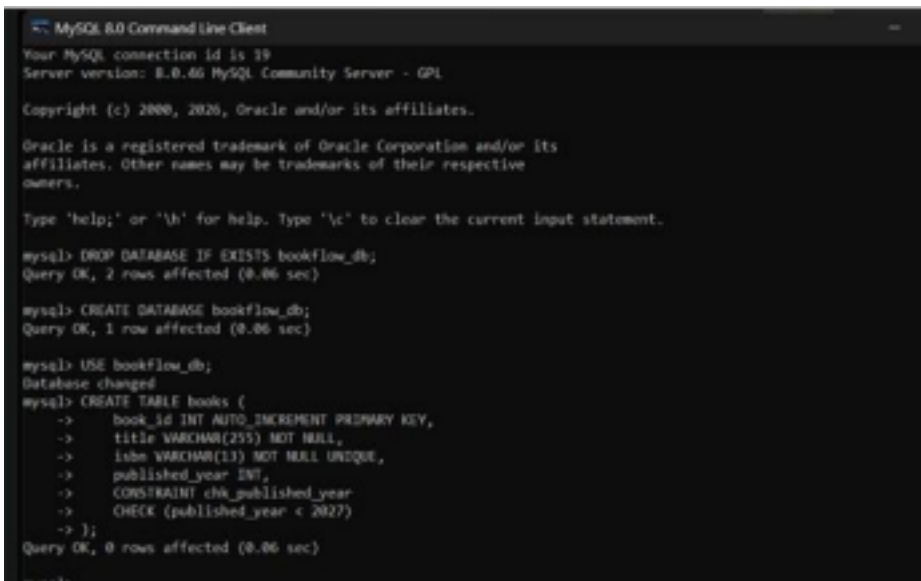
CREATE DATABASE IF NOT EXISTS bookflow_db;



```
mysql> CREATE DATABASE bookflow_db;
ERROR 1007 (Warning): Can't create database 'bookflow_db'; database exists
mysql> USE bookflow_db;
Database changed
mysql> Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_bookflow_db |
+-----+
| books                  |
| members               |
+-----+
2 rows in set (0.02 sec)
```

Query 2: Select the database

USE bookflow_db;



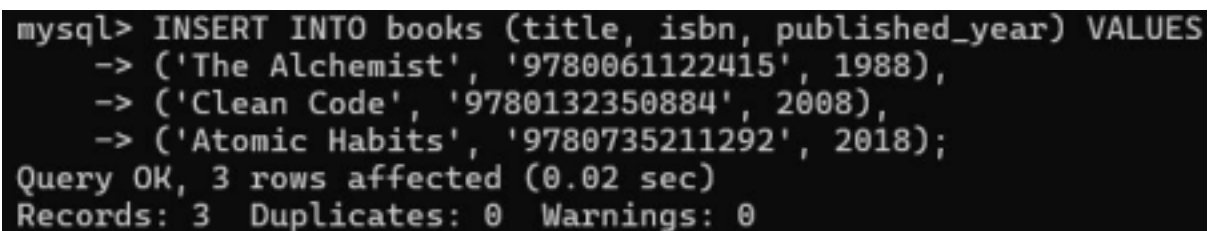
```
mysql> DROP DATABASE IF EXISTS bookflow_db;
Query OK, 2 rows affected (0.06 sec)

mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.06 sec)

mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE books (
  ->   book_id INT AUTO_INCREMENT PRIMARY KEY,
  ->   title VARCHAR(255) NOT NULL,
  ->   isbn VARCHAR(13) NOT NULL UNIQUE,
  ->   published_year INT,
  ->   CONSTRAINT chk_published_year
  ->   CHECK (published_year < 2027)
  -> );
Query OK, 0 rows affected (0.06 sec)
```

3. Step 2 Table Creation (DDL) with Constraints

Query 3: Create the books table



```
mysql> INSERT INTO books (title, isbn, published_year) VALUES
  -> ('The Alchemist', '9780061122415', 1988),
  -> ('Clean Code', '9780132350884', 2008),
  -> ('Atomic Habits', '9780735211292', 2018);
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM books;
```

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

```
3 rows in set (0.00 sec)
```

Query 4: Create the members table

```
mysql> INSERT INTO members (full_name, email) VALUES
-> ('Anil Kumar', 'anil.kumar@example.com'),
-> ('Priya Sharma', 'priya.sharma@example.com'),
-> ('Ravi Verma', 'ravi.verma@example.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM members;
```

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com

```
3 rows in set (0.00 sec)
```

Query 5 (verification): Inspect the table structures

```
mysql> DESCRIBE books;
```

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(255)	NO		NULL	
isbn	varchar(13)	NO	UNI	NULL	
published_year	int	YES		NULL	

```
4 rows in set (0.00 sec)
```

```
mysql> DESCRIBE members;
```

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

```
3 rows in set (0.01 sec)
```

```
mysql> |
```

Query 6: Insert 3 books

```
INSERT INTO books (title, isbn, published_year) VALUES
('The Alchemist', '9780061122415', 1988),
('Clean Code', '9780132350884', 2008),
('Atomic Habits', '9780735211292', 2018);
```

Query 7: Insert 3 members

```
INSERT INTO members (full_name, email) VALUES
('Anil Kumar', 'anil.kumar@example.com'),
('Priya Sharma', 'priya.sharma@example.com'),
('Ravi Verma', 'ravi.verma@example.com');
```

Test 1: Duplicate ISBN (UNIQUE violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Fake Copy', '9780061122415', 2000);
```

Explanation: The ISBN 9780061122415 already belongs to 'The Alchemist'. The UNIQUE constraint on isbn rejects the row, so two physical records can never claim the same ISBN.

Test 2: NULL title (NOT NULL violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```

Explanation: The NOT NULL constraint fires because a book must have a name. The database refuses to store a nameless book, exactly as the requirement demands.

Test 3: Future publication year (CHECK violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Time Traveler', '8888888888888', 2030);
```

Explanation: 2030 fails the condition published_year < 2027, so the named CHECK constraint chk_published_year blocks the insert. Garbage dates cannot enter the table.

Test 4: Duplicate email (UNIQUE violation)

```
INSERT INTO members (full_name, email)
VALUES ('Anil Clone', 'anil.kumar@example.com');
```

Explanation: anil.kumar@example.com is already registered to member_id 1, so the UNIQUE constraint on email rejects the second signup. One email = one account

```
mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES ('Fake Copy', '9780061122415', 2000);
ERROR 1062 (23000): Duplicate entry '9780061122415' for key 'books.isbn'
mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES (NULL, '9999999999999', 2010);
ERROR 1048 (23000): Column 'title' cannot be null
mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES ('Time Traveler', '8888888888888', 2030);
ERROR 3819 (HY000): Check constraint 'chk_published_year' is violated.
mysql> INSERT INTO members (full_name, email)
-> VALUES ('Anil Clone', 'anil.kumar@example.com');
ERROR 1062 (23000): Duplicate entry 'anil.kumar@example.com' for key 'members.email'
```